



Traveling Salesman

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Cours	BOX Certificative
Type	Projet
Relue ?	☐
Année	2024.25 L3 SDN
Langage	
Semestre	L3 S6
à compléter	☐

Interesting approaches

Nearest-Neighbor

Complexity : $O(n^2)$

Concept : lets the salesman choose the nearest unvisited city as his next move

Drawback :

- not very precise
- varies a lot depending on the starting point
- for N cities randomly distributed on a plane, the algorithm on average yields a path 25% longer than the shortest possible path, which can scale up fast
- there exist many specially-arranged city distributions which make the NN algorithm give the worst route

Improvement of the solution : the 2-opt heuristic

- concept : takes the messy route generated by NN and optimizes it by eliminating crossed paths (loops), selects two edges (links between cities) at random, swaps them to reverse the order of the segment between them, and checks if the total distance decreases, if it does, the change is kept
- mechanism: It repeats this process iteratively until no more swaps can improve the route (reaching a local optimum)
- complexity : $O(n^2)$ per iteration, making the overall combo still extremely fast
- impact : fixes the "worst route" and "25% longer" drawbacks of NN, bringing the final path down to roughly 5% or less from the absolute shortest path

Christofide

Complexity : $O(n^4)$

Concept :

1. Find a minimum spanning tree for the problem
2. Create duplicates for every edge to create an Eulerian graph
3. Find an Eulerian tour for this graph

4. Convert to TSP : if a city is visited twice, then create a shortcut from the city before this in the tour to the one after this

At most 1.5 times the optimal

Dynamic

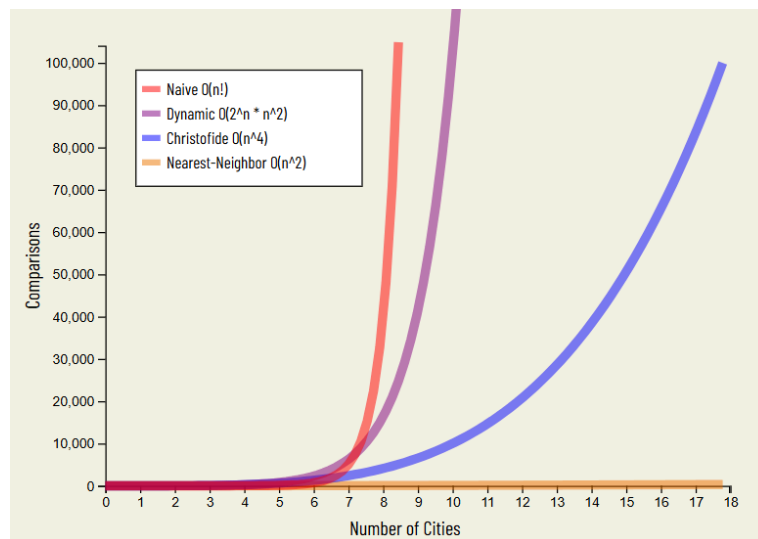
Complexity : $O(2^n * n^2)$

Concept : the input is a distance matrix between a set of cities, and the goal is to find a minimum-length tour that visits each city exactly once before returning to the starting point

Advantage : finds the exact solution of the problem

Drawback : if there is more than 20 cities, the compute time goes from seconds to minutes to hours very fast

Comparison of the approaches



Sources

<https://cse442-17f.github.io/Traveling-Salesman-Algorithms/>

<https://www.lix.polytechnique.fr/page/index.php?username=vjost&path=mpri/TSP.pdf>

https://en.wikipedia.org/wiki/Travelling_salesman_problem

https://en.wikipedia.org/wiki/Held-Karp_algorithm

<https://medium.com/data-science/how-to-solve-the-traveling-salesman-problem-a-comparative-analysis-39056a916c9f>